HDOC Day-58

Name	Gurveer Singh Cheema
SAP ID	590032925
Batch	15

Problem Statement

Write a C program that reads the size and elements of a variable-length integer array **arr[]**, and an integer **k** such that 0 < k <= size. Pass the array to a function named **kth_smallest()** using its address. Use pointers to pass the array between the calling function and **kth_smallest()**.

Arrange the elements of **arr[]** in ascending order using the **bubble sort algorithm**. Find and display the element at the k-th position in the sorted array. Duplicate elements must be counted separately. Display **-1** if **k** is invalid.

C Program

```
#include <stdio.h>

void kth_smallest(int *arr, int size, int k)
{
    int i, j, temp;

    if (k <= 0 || k > size)
    {
        printf("-1\n");
        return;
    }

    /* Bubble sort in ascending order */
    for (i = 0; i < size - 1; i++)
    {
        for (j = 0; j < size - i - 1; j++)
        {
            if (*(arr + j) > *(arr + j + 1))
            {
                temp = *(arr + j);
                *(arr + j) = *(arr + j + 1);
                *(arr + j + 1) = temp;
            }
        }
    }

    printf("kth smallest element: %d\n", *(arr + k - 1));
}

int main()
{
    int size, i, k;

    printf("Enter size of array: ");
    scanf("%d", &size);

    int arr[size];

    printf("Enter array elements: ");
    for (i = 0; i < size; i++)
        scanf("%d", arr + i);

    printf("Enter k: ");
    scanf("%d", &k);

    kth_smallest(arr, size, k);

    return 0;
}
```

Explanation

The array address is passed to **kth_smallest()** through a pointer. The function first validates **k**. It then uses bubble sort to arrange the array in ascending order by comparing and swapping adjacent elements through pointer notation. Since array indexing starts from 0, the k-th smallest value is found at position **k - 1**. Duplicate values remain in the sorted array and are therefore counted separately.

Sample Outputs

```
Example 1
Array: 7 10 4 3 20 15
k: 3
kth smallest element: 7

Example 2
Array: 7 10 4 3 20 15
k: 4
kth smallest element: 10

Example 3
Array: 7 10 4 3 20 15
k: 7
-1

Example 4
Array: 4 2 2 7 5
k: 3
kth smallest element: 4
Sorted array: 2 2 4 5 7
```

Result

Thus, the program successfully finds the k-th smallest element using pointers and bubble sort, counts duplicate elements separately, and displays -1 when k is invalid.